

Google Maps API Key Setup Guide

1. Create or Select a Project

- Go to: <https://console.cloud.google.com>
- Click the Project dropdown in the top bar.
- Click 'New Project'.
- Enter a project name (e.g., FieldTrackingApp).
- Click 'Create'.
- Select the newly created project.

2. Enable Required APIs

- Go to APIs & Services → Library.
- Enable only the APIs you actually need.
- Routes API (for backend routing).
- Maps SDK for Android (if Android app).
- Maps SDK for iOS (if iOS app).
- Maps JavaScript API (if web admin panel).

3. Enable Billing (Mandatory)

- Go to Billing.
- Link a credit/debit card.
- Attach the billing account to your project.
- Without billing enabled, the Routes API will not work.

4. Create API Key

- Go to APIs & Services → Credentials.
- Click 'Create Credentials'.
- Select 'API Key'.
- Google will generate a key (e.g., AlzaSyXXXXXXXXXXXXXX).

5. Restrict the API Key (Very Important)

- Click your newly created API key to edit it.
- Under Application Restrictions:
 - - Backend: Choose IP addresses and add your server IP.
 - - Android: Choose Android apps and add package name + SHA-1 fingerprint.
 - - iOS: Choose iOS apps and add bundle ID.
 - - Web: Choose HTTP referrers and add your domain.
- Under API Restrictions:
 - Select 'Restrict key' and allow only the APIs you enabled (Routes API, Maps SDK, etc.).
 - Never leave the key unrestricted.

6. Recommended Key Setup (Best Practice)

- Create separate keys for different use cases:
 - - Backend Key: Routes API (IP restricted).
 - - Mobile Key: Maps SDK (Android/iOS restricted).
 - - Web Key: JavaScript Maps (HTTP referrer restricted).
- Do not reuse the same key everywhere.

7. Testing & Troubleshooting

- Use Postman or curl to test the Routes API.
- If you receive PERMISSION_DENIED, check:
 - - Billing enabled.
 - - API enabled.
 - - API key restrictions correctly configured.